higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MULTIMEDIA BASICS
NQF LEVEL 2

(10040162)

20 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 11 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Complete the following sentences by writing down the missing word(s). Write only the word(s) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 A client application that fetches and displays web pages and other www. resources to the user is called a …

1.1.2 An Acrobat document or file format containing embedded images, graphics and fonts is called ...

1.1.3 … is the delivery of information that combines different content formats, motion video, audio, still images, graphics, animation and text.

1.1.4 … is an animation technique that is used to gradually turn one image into another.

1.1.5 The process of transforming analogue video signal into digital information is called …

1.1.6 Any technology that converts analogue signals, such as video and voice signals, into digital form and compresses them to conserve bandwidth on a transmission path is called ...

1.1.7 A collection of richly formatted graphic/hypermedia documents located on computers around the world and logically linked together by the internet is called ...

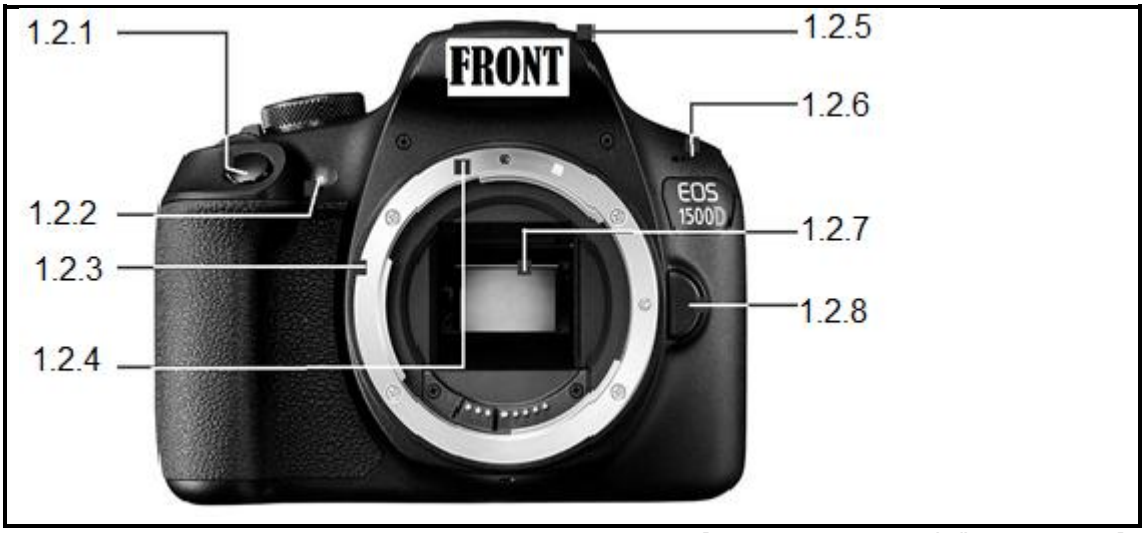
1.1.8 ... allows the user to play media from the internet immediately, without first having to download the entire media file.

1.1.9 … is the most common element of multimedia.

1.1.10 A simulation of movement created by displaying a series of pictures, or frames in cartoons, is an example of …

(10 × 1) (10)

1.2 Study FIGURE 1 and FIGURE 2 below and identify the parts of a digital camera as indicated. Write the part number next to the question number (1.2.1–1.2.8) in the ANSWER BOOK.



[Source: *www.google/images.co.za*]

FIGURE 1



[Source: *www.google/images.co.za*

FIGURE 2

(10 × 1) (10)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.15) in the ANSWER BOOK.

1.3.1 Digital storytelling is using a computer's audio and visual tools to tell a story.

1.3.2 Storyboards are used in storytelling to create a plan and/or layout for a story.

1.3.3 The first step of creating a digital story is to edit it.

1.3.4 The best storytellers are those who give audiences a unique experience.

1.3.5 An over-exposed photograph would be very light.

1.3.6 A good story makes you want to listen, it communicates to your life and shows real situations.

1.3.7 It is a good idea to use pictures when telling stories to children.

1.3.8 The voice is an element of digital storytelling.

1.3.9 The process of reviewing, revising and proof-reading stories is called editing.

1.3.10 Any music could be used to supplement or complement a story.

1.3.11 In newspapers, stories are measured in inches and/or centimetres.

1.3.12 Young people are often most attracted to *realistic* stories.

1.3.13 Transitions and graphics are elements of digital storytelling.

1.3.14 Every good news story must have a beginning, middle and end.

1.3.15 A fictional story tries to tell the truth about the antagonist in the story.

(15 × 1) (15)

1.4 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.4.1–1.4.15) in the ANSWER BOOK.

1.4.1 A sound signal travels faster in ...

A air.
B a liquid.
C solid material.
D a vacuum.

1.4.2 A sound wave of a single frequency is called ...

A music.
B a tone.
C noise.
D distortion.

1.4.3 Distortion occurs in a recording when ...

A too many sound signals are recorded at the same time.
B sound is too intense or too loud and its true quality alters.
C the listeners cannot hear the sound.
D the singers do not practice well.

1.4.4 External noise is defined as …

A noise which cannot be recorded by a sound system.
B noise which is outside of the recording room or studio.
C the type of noise which is generally external due to a sound system.
D internal noise of the sound system.

1.4.5 … is produced when something vibrates.

A Air
B Light
C Sound
D Energy

1.4.6 The following is a type of external noise:

A industrial noise
B electron noise
C thermal noise
D crosstalk

1.4.7 The internal noise of a sound recording system is defined as …

A noise which is currently recorded by a sound system.
B noise outside of the recording room or studio.
C the type of noise which is generally external due to a sound system.
D noise in the internal components of the sound system.

1.4.8 When music is recorded in a room, background noise refer to the …

A ambient noise in a room unrelated to the instrument(s).
B vocal(s) being recorded.
C system noise.
D guitar sound being recorded.

1.4.9 When dubbing is applied to analogue sound, it refers to ...

A repeating a song on a tape.
B erasing a previously recorded tape.
C copying a recording from one tape to the other.
D converting analogue sound on tape to digital sound.

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1.4.10 Crosstalk in sound production means ...

A leakage of sound from one room to the other.
B fading of the current audio signal.
C mixing of two unwanted audio channels.
D the unwanted leakage of an audio signal between two audio channels.

1.4.11 Blending in sound processing can be defined as ...

A the mixing of multiple sounds or channels together to form one sound.
B the blocking of unwanted noise while recording.
C increasing the volume of recorded sound.
D balancing the noise and sound in music.

1.4.12 The threshold of hearing in humans can be defined as ...

A a soft noise that is not audible to humans.
B a general type of noise.
C minimum and maximum range of sounds that could be heard.
D any sound that is audible to humans.

1.4.13 A speaker connected to a music system is ...

A digital equipment that is used to convert bits to sound.
B a device that converts electrical signals to sound.
C an input device also found on TV systems.
D the last part of any music equipment.

1.4.14 Narrowband noise is described as ...

A less important noise in a system.
B noise that occurs over a limited frequency range.
C noise that does not cause much distortion to the signals.
D noise at low frequencies.

1.4.15 Sound processing can be defined as ...

A modulation of unwanted signals
B actual performance by the musician
C making the sound pass through many devices
D the intentional alteration of audio signals

(15 × 1) (15)
[15]

TOTAL SECTION A: 50

SECTION B

QUESTION 2

2.1 Study the poster below and answer the questions.



[Source: *www.google/images.co.za*]

FIGURE 3

2.1.1 Explain the importance of hierarchy in typographic designs. (3)

2.1.2 Discuss the purpose of the first level of typography in a poster. (3)

2.1.3 Identify TWO levels (except the first level as mentioned in QUESTION 2.1.2) which make the above poster attractive. (2 × 2) (4)

2.1.4 Give any THREE advantages of using a poster to communicate a message. (3 × 2) (6)

2.2 Define the term *rule of thirds*. (2)

2.3 Name TWO file formats that could be used to save images. (2 × 1) (2)

[20]

QUESTION 3

3.1 Describe FIVE benefits an advertising company could gain from using animation on their website. (5 × 2) (10)

3.2 Explain the greatest advantage of using GIF (Graphic Interchange Format) as a form of web animation on a website. (2)

3.3 Define the following terms that apply to animation:

3.3.1 Roto-scoping

3.3.2 Motion capture

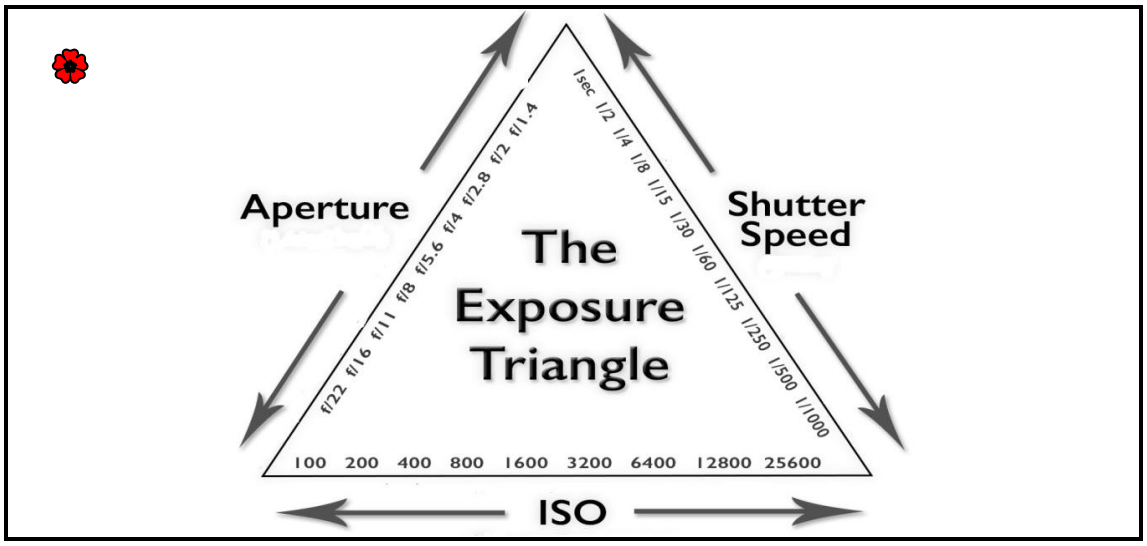
3.3.3 Key-frame animation

3.3.4 Three-point lighting

(4 × 2) (8)
[20]

QUESTION 4

4.1 Study the following diagram that shows the exposure triangle of a digital photographic camera and answer the questions:



[Source: *www.google/images.co.za*]

FIGURE 4

4.1.1 Discuss the importance of the camera's *ISO* setting. (3)

4.1.2 Briefly explain how a photographer could adjust the shutter speed of a camera when photographing a sports person in action. (3)

4.1.3 Describe the role of the aperture setting when taking a photograph. (3)

4.2 Briefly describe any FOUR disadvantages of analogue (film) photography. (4 × 2) (8)

4.3 Describe the working of the digital SLR or DSLR camera. (3)
[20]

QUESTION 5

5.1 List SIX principles of animation which should be observed when creating animation movies. (6 × 2) (12)

5.2 Discuss how the following could be used in 3D animation:

5.2.1 Scan-line rendering

5.2.2 Web animation

5.2.3 Flash animation

5.2.4 Rendering

(4 × 2) (8)
[20]

QUESTION 6

Study the story below and answer the questions.

Once a saint visited a town for some days. During his stay, the saint would pray silently sitting under a tree. People got to know about him and would come to visit him to ask him for solutions to their problems.

In that town also lived a very wealthy trader. After observing the saint for a few days, the trader was deeply touched by the saint's devotion and sincerity.

One day the wealthy man went to the saint and offered him a bag of money saying, 'I know you will use this money for the good of other needy people, so please accept it'.

The saint replied, 'Please wait a moment. I am not sure if it would be right for me to take your money'.

The saint questioned, 'Are you wealthy? Do you have more money at home?'
The trader proudly replied, 'Yes, I have thousands more money coins at home'.

The saint again questioned, 'Do you want thousands of more money coins?'
The trader replied, 'Why not, I work hard every day to earn more money'.

The saint questioned, 'Do you wish to get more thousands of money pieces beyond what you already have?'
The trader replied, 'Certainly, every day I pray that I may earn more and more money ...'

The saint pushed back the bag of money to the trader and said, 'I am sorry but I cannot take your money'.

The trader was surprised. He asked, 'But why …?'

The saint replied, 'A wealthy man cannot take money from a beggar'.

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The wealthy man felt offended and gasped, 'How can you call me a beggar and yourself a wealthy man?'

The saint smiled and replied, 'I am wealthy because I am content with whatever I have and you are a beggar because no matter how much you possess still you are always dissatisfied and begging for more'.

6.1 Write a short synopsis of the story. (5)

6.2 Identify the hero of the story. (1)

6.3 Discuss any THREE characteristics of the story that make it suitable for television. (3 × 2) (6)

6.4 Briefly explain the type of conflict in this story. (2)

6.5 Identify the following characters:

6.5.1 Main character

6.5.2 Incidental character(s)

6.5.3 Protagonist

(3 × 2) (6)
[20]

TOTAL SECTION B: 100
GRAND TOTAL: 150

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